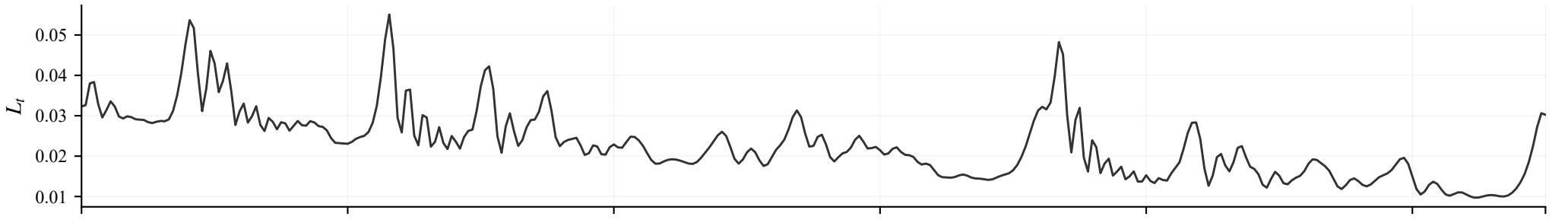
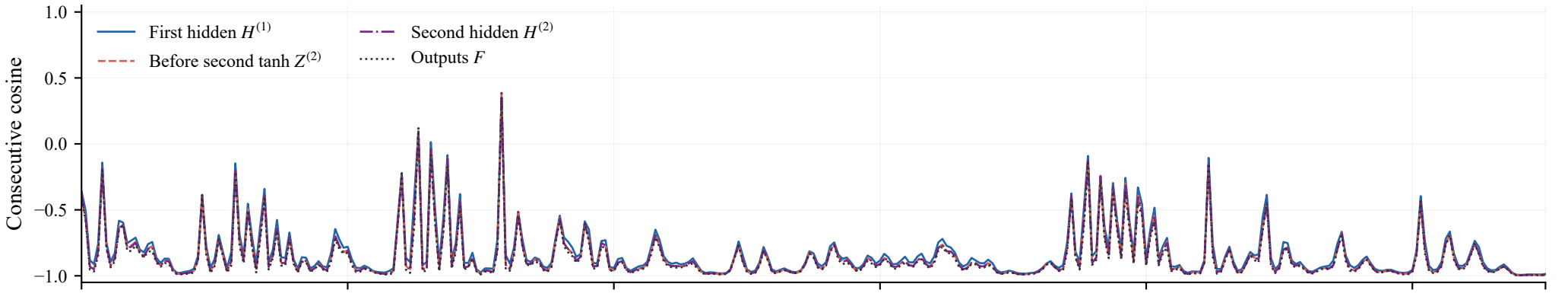
CIFAR-10 MLP: how the first hidden oscillation reaches the second layer

$Z_t^{(2)} = H_t^{(1)} W_{2,t}^T + b_{2,t}, \quad H_t^{(2)} = \tanh(Z_t^{(2)})$; rows are samples, columns are activations.

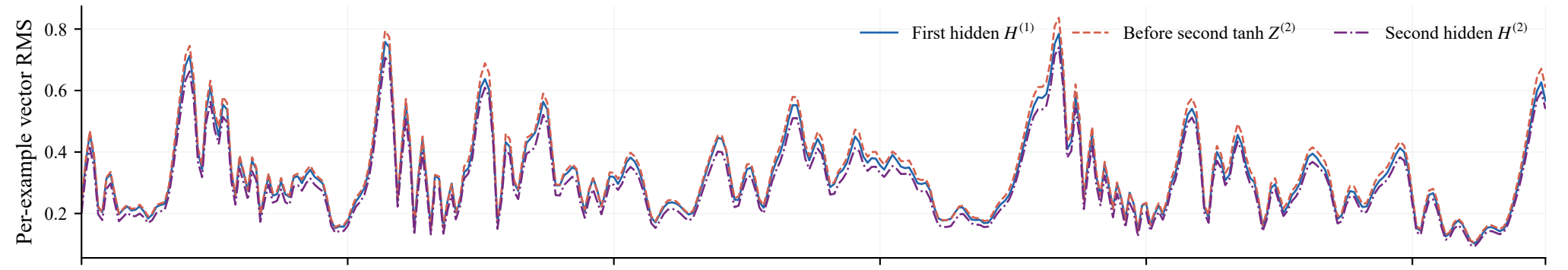
(a) Full training loss



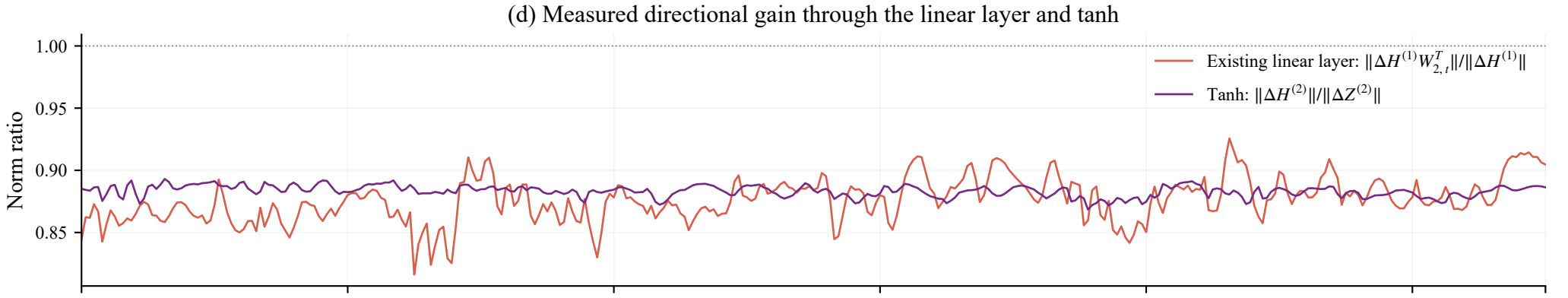
(b) Reversal geometry across the two hidden layers and outputs



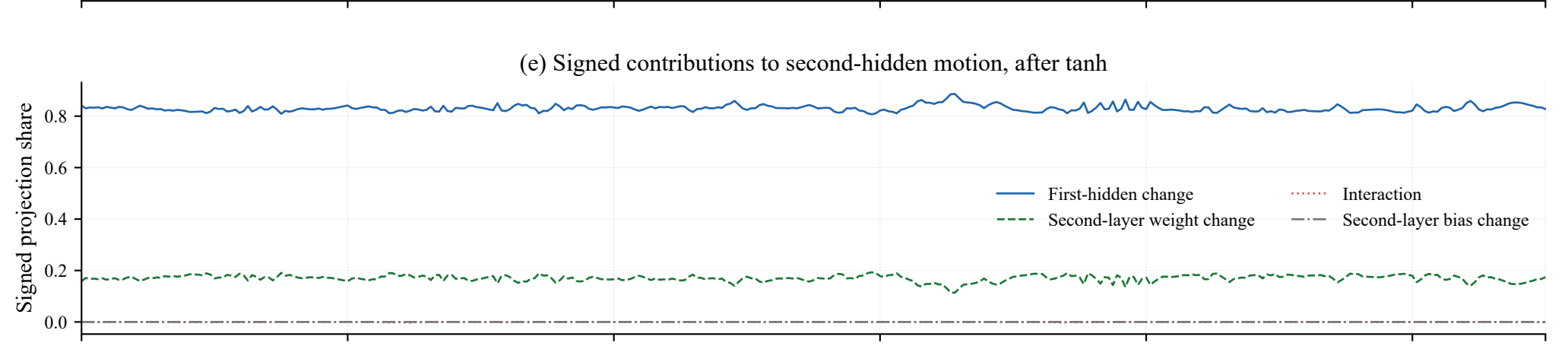
(c) Displacement amplitude along the representation pathway



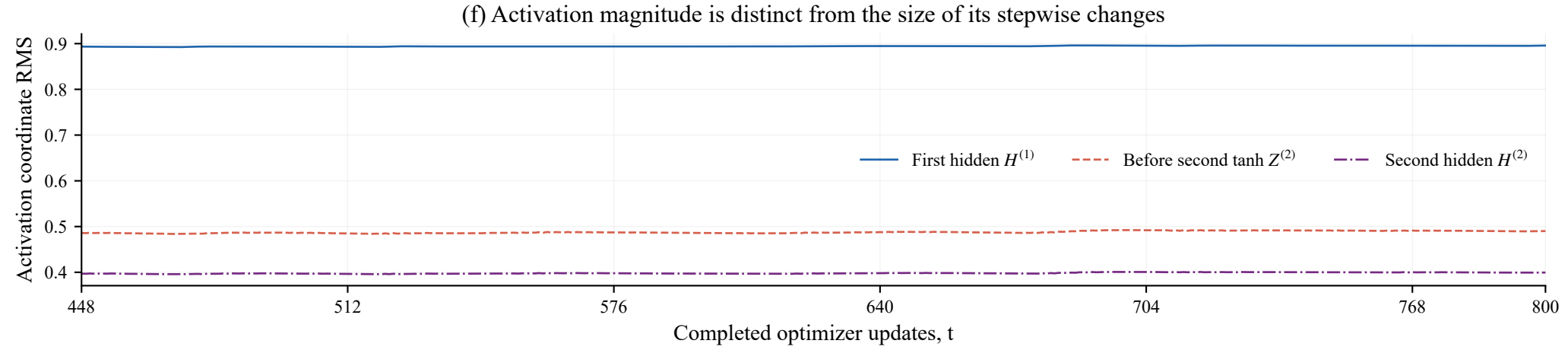
(d) Measured directional gain through the linear layer and tanh



(e) Signed contributions to second-hidden motion, after tanh



(f) Activation magnitude is distinct from the size of its stepwise changes



Panel (e) uses the observed elementwise tanh secant to transmit each additive preactivation contribution.

Signed projection shares can be negative; they sum to one including the retained forward-roundoff residual.

These are measured-trajectory identities, not interventions or an identification of the unique initiating cause.